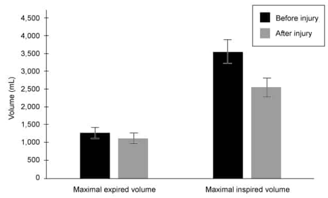
A cyclist's lung function was tested in preparation for a race. During the race, she was struck by another cyclist and sustained an injury that caused impaired diaphragm contraction. The graph below shows lung volume changes from baseline during deep-breathing tests conducted before and after the injury.

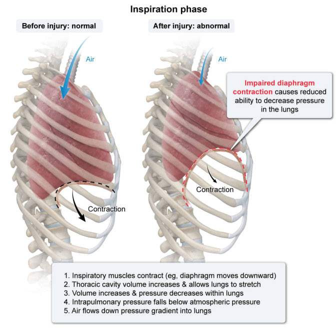


Given the observed changes following the injury, the cyclist would most likely exhibit a reduced ability to:

- A. increase pressure inside the lungs during inspiration. [28%]
- ✔ B. decrease pressure inside the lungs during inspiration. [66%]
- C. increase pressure inside the lungs during expiration. [2%]
- D. decrease pressure inside the lungs during expiration. [2%]

Correct
65% answered correctly

Explanation:



The two phases of breathing are inspiration and expiration. Inspiratory muscles such as the **diaphragm**, a muscular partition beneath the lungs, work in concert with each other to facilitate breathing.

During **inspiration**, inspiratory muscles contract (eg, the **diaphragm** moves downward) to increase the volume of the thoracic cavity. The lungs stretch within this larger space, which leads to increased intrapulmonary volume and an inversely proportional decrease in intrapulmonary pressure. Eventually, the air pressure in the lungs *falls below* atmospheric pressure and, because gases flow down their pressure gradients, air will flow from the higher-pressure environment of the atmosphere into the lower-pressure environment of the lungs. This air flow persists until intrapulmonary pressure equals

atmospheric pressure.

In contrast, during **expiration** inspiratory muscles relax (eg, the diaphragm moves upward) to decrease the volume of the thoracic cavity. The lungs recoil passively to fit this smaller space, which leads to a decrease in intrapulmonary volume and an inversely proportional *increase in intrapulmonary pressure* (until it is *higher than* atmospheric pressure). Subsequently, air flows down its pressure gradient as it is forced out of the higher-pressure environment in the lungs and into the lower-pressure environment of the atmosphere.

In this scenario, a cyclist sustained an injury that resulted in impaired diaphragm contraction during breathing. As the graph shows, this injury had no significant effect on lung volume during maximal expiration but resulted in significantly *reduced lung volume* during *maximal inspiration*. Accordingly, the cyclist is unable to maximally expand the thoracic cavity and lungs during inspiration. This would lead to a **reduced ability** to **decrease pressure in the lungs** during **inspiration**.

(Choice A) Any contraction of the diaphragm during inspiration (even if impaired) would *increase* the volume of the thoracic cavity and cause *decreased* (*not* increased) pressure inside the lungs.

(Choices C and D) Because the cyclist's lung volume during expiration was *not* significantly affected by the injury, her ability to *increase* pressure inside the lungs during expiration is most likely *unaffected*.

Educational objective:
During inspiration the diaphragm contracts, expanding the thoracic cavity and reducing intrapulmonary pressure. This causes the lungs to expand. As lung volume increases, intrapulmonary pressure drops below that of the atmosphere, and air flows from the atmosphere (higher pressure) into the lungs (lower pressure).

Time spent:0 sec
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System:
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